

Zeyu (Derek) Zeng

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Education

Beijing 101 High School

2021.9 - 2024.7

University of Illinois Urbana-Champaign

2024.8 - 2028.5 (Expected)

(Current) GPA: 3.93

B.S. in Physics and Mathematics, expected May 2028

(Current) Technical GPA: 3.91

Selected Courses:

- **Honors Course:** MATH 347H (Introduction to Abstract Mathematics Honor Session, A), MATH 416H (Abstract Linear Algebra Honor Session, A+), MATH 424 (Honors Real Analysis, A+), MATH 428 (Honors Topics Course on Morse Theory).
- **Graduate Courses:** MATH 535 (General Topology, A+), MATH 518 (Differentiable Manifold I, A+), MATH 519 (Differentiable Manifold II, A), MATH 525 (Algebraic Topology I, A), MATH 526 (Algebraic Topology II), MATH 514 (Complex Algebraic Geometry)
- **Fields Institute Shared Graduate Course:** Geometry of Quantum Mechanics (Fall 2025), Mathematics Inspired by Quantum Field Theory (Winter 2025-2026), Introduction to Complex Manifolds (Fall 2026)

Publications and Preprints

Publications:

Zeyu Zeng and Elena Kopteva, Isotropic Coordinates for Generalized Schwarzschild-like Solutions. *J. Math. Phys.* 1 June 2026; 67 (6): 062501. doi:10.1063/5.0335825; arXiv:2603.21040 [gr-qc].

Zeyu Zeng, Baoyi Chen, and Jiaying Zhao, Relativistic Corrections to Hadron-Hadron Scattering Phase Shift and Correlation Function. *Chinese Physics C*, 2026. doi:10.1088/1674-1137/ae5884; arXiv:2506.19240 [hep-ph].

Preprints:

Zeyu Zeng, Single Lie Brackets on Closed Manifolds. arXiv:2609.05958 [math.DG].

Undergraduate Research Experiences

Single Lie Brackets on Closed Manifolds

2026.6 - 2026.9

- This is my independent work on differential geometry.
- Proved that every smooth vector field on a closed manifold is a single Lie bracket, without assuming orientability.
- Developed a geometric construction combining integration along flows with local extensions, using transversality and explicit corrections to make the solutions agree globally.
- arXiv:2609.05958 [math.DG]

Schwarzschild-like Metric in Isotropic Coordinates

2025.3 - 2026.3

- Research mentor: Dr. Elena Koptieva
- Derive the coordinate transformation from the canonical form to the isotropic coordinate.
- Develop a general method for transforming Kiselev solutions into isotropic coordinates
- See arXiv:2603.21040 [gr-qc]

Illinois Math Lab Research Project, Arithmetic of Polynomials in Knot Theory

2025.8 - 2025.12

- Research mentor: Dr. Yi Wang
- Study knot theory by the factorization of a polynomial related to the trace field of $-a/b$ -Dehn filling.
- See the final report [here](#)

Relativistic Corrections to Hadron-Hadron Correlation Function

2023.11 - 2025.6

- Research mentors: Prof. Baoyi Chen, Dr. Jiaying Zhao
- Examine relativistic corrections to scattering phase shifts and correlation functions using the two-body Dirac equation framework.
- See arXiv:2506.19240 [hep-ph]

Talks and Presentations

From 2D TQFT to Gromov-Witten Theory

2026.4

- This presentation was based on McDuff-Salamon's *J-holomorphic Curves and Symplectic Topology* and Pandharipande's ICM 2018 lecture notes: [Cohomological Field Theory Calculations](#).
- The final presentation on Fields Institute Shared Graduate Course: Mathematics Inspired by QFT, Winter 2025-2026

Arithmetic of Polynomials in Knot Theory

2025.12

- The final presentation of the 2025 Fall IML project with the same name.

Gauge Theory of Shallow Water

2025.7

- "The Largest and The Smallest", SJTU Summer School on Physics, final presentation with collaborators.
- The presentation is mainly based on David Tong's paper [arXiv:2209.10574 \[hep-th\]](#).

Derivation of Conformally Flat Metric for Generalized Schwarzschild-like Solutions

2025.4

- UIUC Undergraduate Research Symposium, Poster Presentation.
- Contains the basic notion of isotropic coordinates and part of the idea of the research project with the same name.

Supervised Readings

Heegaard-Floer and Bordered Floer Theory

Fall 2026

- Supervisor: Prof. Jake Rasmussen
- The reading course is mainly based on the note [An Introduction to Heegaard Floer Homology](#) by Peter Ozsváth and Zoltán Szabó.

Stable Homotopy Theory

Fall 2026

- Supervisor: Prof. Charles Rezk
- The reading is mainly based on the book [Foundations of Stable Homotopy Theory](#) by David Barnes and Constanze Roitzheim.

Directed Reading Program: Graded Differential Geometry

Fall 2026

- Supervisor: Nikolay Sheshko
- The first part of the reading will be based on [Lecture Notes of Symplectic Geometry of Graded Manifolds and Higher Lie Groupoids](#); other reading material includes [Derived Differentiable Manifolds](#) and [Orbifolds and Stacks?](#).

Simplicial Homotopy Theory and ∞ -Categories

Summer 2026

- Supervisor: Prof. Charles Rezk
- The first part of the reading is based on [An Elementary Illustrated Introduction to Simplicial Sets](#) by Greg Friedman, [arXiv 0809.4221](#). The second part is based on [Simplicial Homotopy Theory](#) by Edward Curtis and the book [Simplicial Objects in Algebraic Topology](#) by Peter May. The last part is about higher categories, based on [Rezk's note](#), the book [Introduction to Infinity-Categories](#) by Markus Land, and the [Kerodon](#) webpage.
- In the first and second parts about simplicial homotopy theory, I have learned the basic definition of simplicial sets; homotopy groups via simplicial methods, lifting properties, Moore-Postnikov systems, simplicial (abelian) groups and the Dold-Kan correspondence, simplicial methods on fiber bundles and classifying spaces.
- In the last part of the reading, I have learned the basic language of infinity categories, including the simplicial definition of the infinity category and functors between them, the homotopy category and the nerve of 1-categories, analogies of 1-category theory in infinity categories (joints, slices, limits, colimits, categorical equivalences), Joyal's extension and lifting theorems.

Differential Topology and Cobordism Theory

Spring 2026

- Supervisor: Prof. Dan Berwick-Evans
- Read Milnor's *Topology from the Differentiable Viewpoint*, Lectures 1–7 of Freed's [Bordism: Old and New](#), and Chapters 1–5 of Milnor-Stasheff's *Characteristic Classes*.
- Studied foundational topics in differential topology, including Sard's theorem, regular values, degree theory, the Poincaré-Hopf theorem, and the Pontryagin-Thom construction of framed cobordism.
- Also learned the basic definitions of the cobordism ring, stable framing, and the intrinsic definition of the stable Pontryagin-Thom construction, applications of the Pontryagin-Thom construction in the computation of stable homotopy groups, and introductory constructions of characteristic classes.
- [Here](#) is my reading report.

Directed Reading Program: Symplectic Geometry

Spring 2026

- Supervisor: Nikolay Sheshko
- Read the book *Lectures on Symplectic Geometry* by da Silva.

- Learned some fundamental results of symplectic geometry, including generating functions of symplectomorphisms, Darboux’s theorem, Weinstein’s tubular neighborhood theorem, contact structures, and basic complex geometry.

Expository Writings and Study Notes

For a complete list, see [this page](#).

Honors

2026 Selma Wanna Memorial Award	2026.6
Dean’s List, UIUC Grainger College of Engineering	Fall 2024, Fall 2025, Spring 2026
Ralph O. Simmons Undergraduate Research Scholarship	2025.4
UIUC Edmund J. James Scholar	2025.2 - present

Academic Programs and Workshops

Reading Group on Noncommutative Geometry, Organizer	2026.8-2026.12
• This is the homepage for the reading group.	
Reading Seminar on Cobordism and Floer Theory, Organizer	2026.12-2027.1
• This is the homepage for the reading group.	
BICMR 2026 Summer School on Differential Geometry	2026.8
Reading Seminar on Characteristic Classes, Organizer	2026.5-2026.8
• This is the homepage for the reading group.	
“The Largest and the Smallest”, 2025 SJTU Summer School on Physics	2025.7